

# Rahul Yadav

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## EXPERIENCE

**Oracle** Aug 2024 – Present  
*Senior Data Scientist (AI/ML Engineering) | Construction Intelligence Team* Malvern, PA

**ConstructIQ: Multi-Agent AI Platform for Oracle Construction Cloud**

- Architected 8-agent platform unifying 5 data modalities (P6, ERP, contracts/BIM, CCTV, audio) for Oracle Construction Cloud
- Built NEXUS orchestrator on OpenAI Agents SDK coordinating 7 specialist sub-agents over Apache AGE knowledge graph
- Engineered audio NLP pipeline (Whisper → NER → Claude action classification) automating inspection-to-action workflow
- Built RAG agent on Oracle 23ai DB across contracts, RFIs, and BIM with hybrid BM25 + vector search at <200ms P95 latency
- Built Kafka consumers normalising vision events into NEXUS knowledge graph, correlated against P6 activity baseline
- Established LangSmith LLM-as-judge eval harness gating CI/CD across all 8 agents on quality, latency, and cost
- Shipped to 3 design-partner pilots; early results: ~15% fewer overruns and reporting cut from 40 → 8 hrs/month

**Hilabs** Feb 2024 – May 2024  
*Senior Data Scientist | Applied Data Science Team* Washington, D.C.

**LLM-Powered Contract Intelligence for Healthcare Claim Adjudication**

- Delivered document understanding system extracting parties, terms, and pricing rules across 5+ insurance contract types
- Achieved 87% F1 on entity extraction by building custom LayoutLM pipeline for joint text/table/image processing
- Built LangChain indexing over OpenSearch with hybrid metadata + semantic retrieval, supporting downstream claim agents
- Deployed self-hosted fine-tuned Mistral with SQS async processing on private VPC, meeting HIPAA and PHI requirements
- Delivered POCs to 3 enterprise insurance clients, validating pricing accuracy and fit with their claim processing applications

**Apexanalytix** May 2023 – Feb 2024  
*Data Scientist (Intern → FTE) | Analytics Team* Remote, New York

**Enterprise RAG Assistant for Internal & Client Knowledge Access**

- Achieved 89% human-eval approval by building production RAG chatbot on LangChain, ChromaDB, and OpenAI generation
- Lifted retrieval relevance by 37% over flat-chunk baseline on internal eval set via parent-child chunking with MMR scoring
- Built feedback collection pipeline driving iterative tuning of prompts, retrieval, and reranking based on real user query patterns
- Drove adoption across 4 internal and 5 external client applications, replacing manual document search across the org

**Oracle** Sep 2020 – Aug 2022  
*Senior ML Engineer | Oracle Financial ML Team* Bengaluru, India

**Preemptive Anomaly Detection for Corporate Billing (ORMB)**

- Hit 92% precision on 1.2M bills and 5M segments in 20 min via in-memory multivariate detection on Oracle in-database ML
- Built semi-supervised classifier with local + global model explainability for analyst-facing audit and override workflows
- Shipped into Oracle ORMB, deployed across banking, insurance, and telco customers running the platform globally
- Inventor on US Patent 12,333,585 [US12333585B2] for anomaly detection methodology applied to enterprise billing

**Wipro Limited** Jun 2018 – Sep 2020  
*AI/ML Engineer | AI Research Team* Bengaluru, India

**NLP Ticketing Assistant for Employee Support Operations**

- Cut human-agent intervention by 74% and wait time from 18 → 4 min via BERT-powered semantic search over ticket corpus
- Hit 79% accuracy across 10+ ticket categories via intent classifier and resolution retrieval, powering auto-routing in production

## PATENTS & PUBLICATIONS

5 Granted · 4 Pending · 1 Paper

*Inventor on 9 USPTO filings spanning anomaly detection, NLP topic categorization, document AI, and multimodal emotion recognition.*

- **Granted:** US 11,386,712 — Multimodal Analysis Based Emotion Recognition [link]
- **Granted:** US 12,002,091 — Technology System for Assisting Financial Institutions in Debt Collection [link]
- **Granted:** US 12,333,585 — Anomaly Detection for Bill Generation (productized in Oracle ORMB) [link]
- **Granted:** US 12,443,792 — Reference Driven NLP-Based Topic Categorization [link]
- **Granted:** US 12,340,613 — Framework for Identifying Documents [link]
- **Pending:** US App. 17/851,267 — ML Pipeline with Data Quality Framework [link]
- **Pending:** US App. 18/056,456 — Predicting Record Topic Using Transitive Relations [link]
- **Pending:** US App. 17/989,202 — Hierarchical Visualization of Clustered Datasets [link]
- **Pending:** US App. 19/320,494 — Reference Driven NLP-Based Topic Categorization(continuation of US 12,443,792) [link]
- **Publication:** Virtual Conversation with Real-Time Body Movement / Gesture Prediction — ICMLIP 2019 [link]

## EDUCATION

### University at Buffalo, SUNY

Aug. 2022 – Dec. 2023

*Master of Science, Computer Science and Engineering – CGPA 3.82 / 4.0*

Buffalo, NY, USA

- Coursework: NLP, Machine Learning, Pattern Recognition, Deep Learning, Reinforcement Learning, Computing Systems

### Indian Institute of Technology (BHU), Varanasi

Jul. 2014 – May 2018

*Bachelor of Technology, Electronics Engineering*

Varanasi, India

## PROJECTS

### A2A + MCP Multi-Agent Workflow [\[code\]](#)

2025 – 2026

- Multi-agent system on OpenAI Agents SDK using Agent-to-Agent (A2A) communication and Model Context Protocol (MCP) for standardized tool and context sharing
- Structured handoffs across agents covering task decomposition, parallel execution, and result synthesis as a reference for production A2A patterns

### Dynamic Agent Spawning Framework [\[code\]](#)

2026

- Runtime agent creation system where LLM dynamically decides agent type, tool set, and prompts per request, no static roster
- Scales agent count and complexity to task difficulty, cutting idle compute vs fixed multi-agent fleets on lightweight queries

### Code Review Agent [\[code\]](#)

2025 – 2026

- LLM agent combining AST-based static analysis with narrative feedback, ranking severity across security, complexity and tests
- Reviews PRs end-to-end with structured output and inline suggestions, integrating into GitHub workflows

### Network-Based Intrusion Detection System (NIDS) [\[code\]](#) [\[results\]](#)

Research Assistant, Dr. Hongxin Hu | Spring 2023

- Engineered intrusion detection system using deep neural detectors for real-time identification of network security threats
- Benchmarked 4 deep neural detector architectures across 12 attack datasets; identified detection limits per attack type

### From RAG POC to Production: What Actually Changes [\[blog\]](#)

Technical Blog · 2026

- Lessons from shipping production RAG: hybrid retrieval beats vector-only, hallucinations are a product problem not a model problem, observability must be built before launch, and trust is won in the first thousand answers
- Covers tiered model selection (50% cost cut at equivalent quality), semantic caching with 30–50% hit rates, citation-first UX patterns, and per-stage observability for retrieval, faithfulness, and latency

## SKILLS

**Agentic & LLM Systems** : OpenAI Agents SDK, LangGraph, LangChain, LlamaIndex, MCP, A2A, RAG, GraphRAG, Tool-use Agents, Function Calling, Prompt Engineering, LLM-as-Judge, LoRA / QLoRA Fine-tuning, LangSmith, Evaluation Harnesses

**ML & NLP** : PyTorch, TensorFlow, Hugging Face, scikit-learn, Whisper, LayoutLM, Sentence Transformers, Embeddings & Re-rankers, Quantization & Distillation, MLflow, Optuna

**Vision** : YOLOv11, OpenCV, CLIP, GPT-5.4 Vision, RTSP Processing, Object/PPE Detection, Scene Understanding, Segmentation

**Data, Vector & Streaming** : Kafka, PySpark, Snowflake, SQL, MongoDB, Oracle 23ai Vector DB, Weaviate, FAISS, Pinecone, Elasticsearch, Apache AGE, Neo4j

**Cloud, MLOps & Backend** : OCI, AWS, Azure, GCP, Kubernetes, Docker, Terraform, FastAPI, CI/CD, Prometheus, WebSockets

**Programming Languages** : Python, SQL, JavaScript, Java